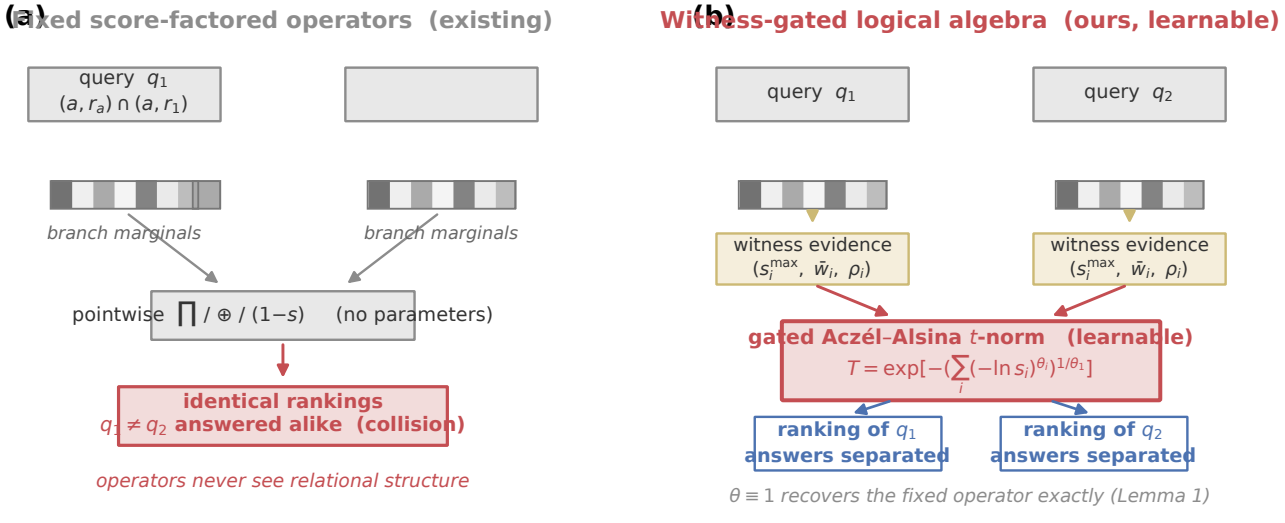
(a)
Fixed score-factored operators (existing)
query q_1
$(a, r_a) \cap (a, r_1)$
branch marginals
branch marginals
pointwise $\prod$ / $\oplus$ / $(1-s)$ (no parameters)
identical rankings
$q_1 \neq q_2$ answered alike (collision)
operators never see relational structure
(b)
Witness-gated logical algebra (ours, learnable)
query q_1
query q_2
witness evidence
$(s_i^{\max}, \bar{w}_i, \rho_i)$
witness evidence
$(s_i^{\max}, \bar{w}_i, \rho_i)$
gated Aczél–Alsina t-norm (learnable)
$T = \exp[-(\sum_i (-\ln s_i)^{\theta_i})^{1/\theta_1}]$
ranking of q_1 answers separated
ranking of q_2 answers separated
$\theta \equiv 1$ recovers the fixed operator exactly (Lemma 1)